

Figure 6 | Software workflow and reproducibility

A CLI/API workflow

snra validate-manifest	checks required fields
snra audit	runs matched controls
snra report	writes audit card and ledger
snra figures	exports source-backed figures

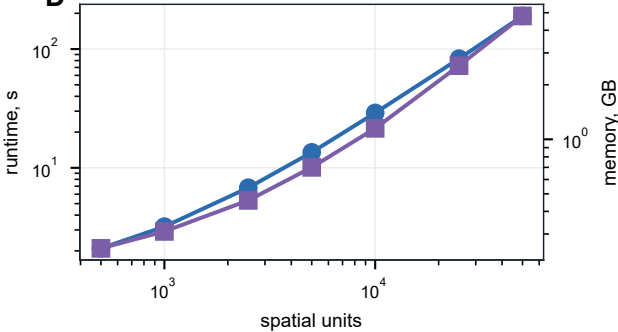
B Input-output schema

input: unit_id, x, y, label, sample
manifest: claim_type, controls, threshold
output: certificate, CI, verdict, caveat
trace: source_data, code, seed, version

C Audit-report example

SNRA audit card	
claim	lineage boundary robustness
certificate	0.881
matched controls	within-sample; within-type
decision	allowed with scope limits

D Runtime/memory scaling



E GitHub documentation/tutorial structure

README: install + quickstart
tutorial: example manifest
API docs: audit objects
examples: source tables + audit cards

F Reproducibility checklist

PASS	package import
PASS	CLI help
PASS	core manifest
PASS	regression tests